



Maths By Gagan Pratap

Trigonometry Sheet-4

Maths Special Batch
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1. Find the value of the following expression.

निम्नलिखित व्यंजक का मान ज्ञात कीजिए। (CDS 2023)

$$5(\sin^4\theta + \cos^4\theta) + 3(\sin^6\theta + \cos^6\theta) + 19 \sin^2\theta \cos^2\theta$$

- (a) 8
(b) 5
(c) 6
(d) 7

2. Find the value of $(\sin^6 A + \cos^6 A + 3\sin^2 A \cos^2 A) - (\sin^4 A - \cos^4 A + 1)\operatorname{cosec}^2 A$?

$(\sin^6 A + \cos^6 A + 3\sin^2 A \cos^2 A) - (\sin^4 A - \cos^4 A + 1)\operatorname{cosec}^2 A$ का मान ज्ञात कीजिये?

(SSC CHSL MAINS 2024)

- A) -3
B) 3
C) -1
D) 0

3. If $f(p) = \sin^{p+2}x + \cos^{p+2}x$, then find $6f(2) - 4f(4) + 10f(0)$?

यदि $f(p) = \sin^{p+2}x + \cos^{p+2}x$, तो $6f(2) - 4f(4) + 10f(0)$ ज्ञात कीजिए? (SSC CHSL PRE 2024)

- A) 14
B) 10
C) 12
D) 11

4. The expression $(\cos^6\theta + \sin^6\theta - 1)(\tan^2\theta + \cot^2\theta + 2) + 3$ is equal to :

व्यंजक $(\cos^6\theta + \sin^6\theta - 1)(\tan^2\theta + \cot^2\theta + 2) + 3$ का मान ज्ञात कीजिए।

- (a) 0
(b) 1
(c) 2
(d) -1

5. $(\operatorname{cosec} A - \sin A)^2 + (\sec A - \cos A)^2 - (\cot A - \tan A)^2$ is equal to:

- (a) 2 (b) 1 (c) 0 (d) -1

6. If $p \sin A - \cos A = 1$, then $p^2 - (1+p^2) \cos A$ equals

यदि $p \sin A - \cos A = 1$, तो $p^2 - (1+p^2) \cos A$ का मान ज्ञात कीजिए।

- (a) 1 (b) -1
(c) 2 (d) 0

7. What is $(\sec^2\alpha + \tan\alpha \cdot \tan\beta - \tan^2\alpha)^2 + (\tan\alpha - \tan\beta)^2 - \sec^2\alpha \cdot \sec^2\beta$ equal to?

$(\sec^2\alpha + \tan\alpha \cdot \tan\beta - \tan^2\alpha)^2 + (\tan\alpha - \tan\beta)^2 - \sec^2\alpha \cdot \sec^2\beta$ किसके बराबर है?

- A) -1 C) 0
B) 1 D) 2

8. If $A + B + C = 90^\circ$, then $\sin \frac{A}{2} \cdot \sin \left(\frac{180-B-C}{2} \right) + \cos \frac{A}{2} \cdot \sin \frac{B+C}{2} = ?$

- a) $\frac{1}{2}$ b) $\frac{1}{\sqrt{2}}$ c) 0 d) $\frac{3}{\sqrt{2}}$

9. Simplify the given expression.

दिए गए व्यंजक का सरलीकरण कीजिए।

$$\frac{1 + \sin^4\theta + \cos^4\theta}{\cos^2\theta + \sin^4\theta}$$

- (a) 3 (b) 1 (c) 2 (d) 4

(SSC CGL 2023 PRE)

10. What is simplified value of $1 + \tan A \cdot \tan \left(\frac{A}{2} \right)$?

- a) $\cos A$ b) $\sin A$ c) $\sec A$ d) $\cot A$



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11. The value of $(2\cos^2\theta - 1) \left(\frac{1+\tan\theta}{1-\tan\theta} + \frac{1-\tan\theta}{1+\tan\theta} \right)$?
- a) 2 b) 4 c) 3 d) 1

12. The value of $\operatorname{cosec}\theta \left[\frac{1+\cos\theta}{\sin\theta} + \frac{\sin\theta}{1+\cos\theta} \right] - 2\cot^2\theta$ is equal to:

$\operatorname{cosec}\theta \left[\frac{1+\cos\theta}{\sin\theta} + \frac{\sin\theta}{1+\cos\theta} \right] - 2\cot^2\theta$ का मान _____ होगा।

SSC Selection Post (Phase-XII)

- [A] 1 [B] 3
[C] 0 [D] 2

13. Solve: $\cos\theta \left[\cot^2\frac{\theta}{2} - \tan^2\frac{\theta}{2} \right] - 4\operatorname{cosec}^2\theta$

- (a) -2 (b) -1
(c) -4 (d) 1

14. $\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$ is equal to:

$\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$ बराबर है:

- a) $1 - \sin A \cos A$ b) $1 + \sin A - \cos A$
c) $1 + \cos A - \sin A$ d) $1 + \sin A \cos A$

15. $\frac{\sin\theta}{\cot\theta + \operatorname{cosec}\theta} - \frac{\sin\theta}{\cot\theta - \operatorname{cosec}\theta} = ?$

- (a) 1 (b) 2
(c) 3 (d) 0

16. $\frac{(\operatorname{cosec}\theta - \sec\theta)(\cot\theta - \tan\theta)}{(\operatorname{cosec}\theta + \sec\theta)(\sec\theta \operatorname{cosec}\theta - 2)} = ?$

- a) 2 b) 1 c) 3 d) -1

17. For $0^\circ < \theta < 90^\circ$, if $\frac{\sec\theta(1-\sin\theta)(\sec\theta+\tan\theta)}{(\sec\theta-\tan\theta)^2} = \frac{1+k}{1-k}$, then k is equal to:

$0^\circ < \theta < 90^\circ$ के लिए, अगर $\frac{\sec\theta(1-\sin\theta)(\sec\theta+\tan\theta)}{(\sec\theta-\tan\theta)^2} = \frac{1+k}{1-k}$ है तो k बराबर है:

- A) $\operatorname{cosec}\theta$ B) $\cos\theta$ C) $\sec\theta$ D) $\sin\theta$

18. $\cos^2\theta + \cos^2(\alpha+\theta) - 2\cos\alpha \cdot \cos\theta \cos(\theta+\alpha) = ?$

- a) $\sin^2\alpha$ b) $\cos^2\alpha$ c) $\tan^2\alpha$ d) $\sec^2\alpha$

19. $\cos^2(A-B) + \cos^2 B - 2\cos(A-B) \cdot \cos A \cdot \cos B = ?$

- a) $\cos^2 A$ b) $\sin^2 A$ c) $\tan^2 A$ d) $\cot^2 A$

20. If $0^\circ < \theta < 90^\circ$, then $\left(\frac{5\cos\theta-4}{3-5\sin\theta} - \frac{3+5\sin\theta}{4+5\cos\theta} \right)$ is equal to

- a) 0 b) 1 c) $\frac{1}{2}$ d) $\frac{1}{4}$

21. Simplify $\frac{1+\sin t}{4-4\sin t} - \frac{1-\sin t}{4+4\sin t}$?

$\frac{1+\sin t}{4-4\sin t} - \frac{1-\sin t}{4+4\sin t}$ को सरल कीजिए?

- A) $4\tan t \cdot \sin t$
B) $\tan t \cdot \sec t$
C) $\tan t - \sin t$
D) $\tan t + \sin t$

22. $\frac{1-\sin A \cos A}{\cos A (\sec A - \operatorname{cosec} A)} \cdot \frac{\sin^2 A - \cos^2 A}{\sin^3 A + \cos^3 A} = ?$

- a) $\sin A$ b) $\cos A$ c) $\tan A$ d) $\operatorname{cosec} A$

23. What is the value of

$$(1 + \cot A + \tan A) (\sin A - \cos A) \frac{\sin A \cos A}{\sin^3 A - \cos^3 A}$$

$(1 + \cot A + \tan A) (\sin A - \cos A) \frac{\sin A \cos A}{\sin^3 A - \cos^3 A}$ का मान क्या है-

- (a) 2
(b) -1
(c) 1
(d) 0

24. $1 - \frac{\sin^2 y}{1+\cos y} + \frac{1+\cos y}{\sin y} - \frac{\sin y}{1-\cos y} = ?$

- a) 0 b) 1 c) $\sin y$ d) $\cos y$

25. The value of $\frac{\sin A}{\cot A + \operatorname{cosec} A} - \frac{\sin A}{\cot A - \operatorname{cosec} A} - 1$ is:



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$$\frac{\sin A}{\cot A + \operatorname{cosec} A} - \frac{\sin A}{\cot A - \operatorname{cosec} A} - 1 \text{ का मान ज्ञात करें।}$$

- (a) $\frac{1}{2}$ (b) 3 (c) 1 (d) 2

26. $\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1}$ is equal to?

- a) $\frac{\sin x - 1}{\cos x}$ b) $\frac{\sin x + 1}{\cos x}$ c) $\frac{\sin x - 1}{\cos x + 1}$ d) $\frac{\sin x + 1}{\cos x + 1}$

27. What is $\left(\frac{\cos \theta - \sin \theta + 1}{\cos \theta + \sin \theta - 1}\right)(\cot \theta - \operatorname{cosec} \theta)$ equal to?

$$\left(\frac{\cos \theta - \sin \theta + 1}{\cos \theta + \sin \theta - 1}\right)(\cot \theta - \operatorname{cosec} \theta) \text{ किसके बराबर है? (UPSC CDS-2 2024)}$$

- A) -1
B) 0
C) 1
D) 2

28. If $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = p \sec \theta + q \tan \theta$, where $0 < \theta < \frac{\pi}{2}$, then what is the value of p+q?

$$\text{यदि } \frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = p \sec \theta + q \tan \theta, \text{ where } 0 < \theta < \frac{\pi}{2}, \text{ जहाँ } 0 < \theta < \frac{\pi}{2} \text{ है, तो } p+q \text{ का मान क्या है?}$$

(CDS-1 2024)

- A) 0
B) 1
C) 2
D) 4

29. Simplify $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}$.

$$\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} \text{ को हल करें:}$$

- (a) $\cos \frac{\theta}{2}$ (b) $\sin \frac{\theta}{2}$
(c) $\cot \frac{\theta}{2}$ (d) $\tan \frac{\theta}{2}$

30. If $x = \frac{2 \sin \theta}{(1 + \cos \theta + \sin \theta)}$, then the value of $\frac{1 - \cos \theta + \sin \theta}{(1 + \sin \theta)}$ is:

$$\text{यदि } x = \frac{2 \sin \theta}{(1 + \cos \theta + \sin \theta)} \text{ है, तो } \frac{1 - \cos \theta + \sin \theta}{(1 + \sin \theta)} \text{ का मान है: (CPO 2022)}$$

- (a) $\frac{x}{(1+x)}$ (b) x (c) $\frac{1}{x}$ (d) $(1+x)$

31. If $x = \frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta}$, then find $\frac{\sin \theta + \cos \theta - 1}{\cos \theta}$?

$$\text{यदि } x = \frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta} \text{ है, तो } \frac{\sin \theta + \cos \theta - 1}{\cos \theta} \text{ ज्ञात कीजिये? (CDS 2022)}$$

- A) $\frac{1}{x}$ C) x
B) $1+x$ D) $x-1$

32. $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} - \frac{2}{\sin 2A} = ?$

- a) -1 b) 1 c) 0 d) 2

33. $[(\sec x \sec y + \tan x \tan y)^2 - (\sec x \tan y + \tan x \sec y)^2] = ?$

- a) -1 b) 0 c) $\sec^2 x$ d) 1

34. The expression $\sqrt{\sin^4 x + 4 \cos^2 x} - \sqrt{\cos^4 x + 4 \sin^2 x}$ is equal to:

$$\sqrt{\sin^4 x + 4 \cos^2 x} - \sqrt{\cos^4 x + 4 \sin^2 x} \text{ का मान ज्ञात करें?}$$

- (a) $\sin 2x$ (b) 1 (c) $\cos 2x$ (d) $-\cos 2x$

35. What is $(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2$ equal to?

$$(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2 \text{ किसके बराबर है? (UPSC CDS-2 2024)}$$

- A) $\tan^2 \alpha \tan^2 \beta$
B) $\sec^2 \alpha \sec^2 \beta$
C) $\tan^2 \alpha \cot^2 \beta$
D) $\sec^2 \alpha \tan^2 \beta$

36. If $\operatorname{cosec} \theta - \sin \theta = p^3$ and $\sec \theta - \cos \theta = q^3$, then what is the value of $\tan \theta$?

$$\text{यदि } \operatorname{cosec} \theta - \sin \theta = p^3 \text{ और } \sec \theta - \cos \theta = q^3 \text{ है तो } \tan \theta \text{ का मान क्या है?}$$



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- (a) $\frac{p}{q}$ (b) $\frac{q}{p}$
(c) pq (d) $p + q$

37. $\frac{\tan A}{1-\cot A} + \frac{\cot A}{1-\tan A} = K + \tan A + \cot A$, then $K = ?$

- (a) 1 (b) 2 (c) 3 (d) 0

38. If $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = k + \tan^2 \theta + \cot^2 \theta$, then the value of k is equal to:

यदि $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = k + \tan^2 \theta + \cot^2 \theta$ है, तो k का मान ज्ञात कीजिए।

- (a) 7 (b) 2 (c) 9 (d) 5

39. If $2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$, then find a, b ?

यदि $\frac{2\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$ है, तो a, b ज्ञात कीजिये?

- A) $-3/2, -1/2$
B) $3/2, 1/2$
C) $-3, -1$
D) $3, 1$

40. If A, B and C are angles of a triangle, then $\sin^2 A + \sin^2 B + \sin^2 C - 2 \cos A \cos B \cos C = ?$

- a) 0 b) 1 c) 2 d) 4

41. If $\sin(X-Y) = \sin X \cos Y - \cos X \sin Y$, then find $\frac{\sin(A-B)}{\cos A \cos B} + \frac{\sin(B-C)}{\cos B \cos C} = ?$

यदि $\sin(X-Y) = \sin X \cos Y - \cos X \sin Y$ है, तो $\frac{\sin(A-B)}{\cos A \cos B} + \frac{\sin(B-C)}{\cos B \cos C}$ ज्ञात कीजिये? SSC CHSL PRE 2024

- A) $\frac{\sin(C-A)}{\cos A \cos C}$
B) $\frac{\sin(A-C)}{\cos A \cos C}$
C) $\frac{\cos(A+C)}{\cos A \cos C}$
D) $\frac{\cos(A+C)}{\sin A \sin C}$

42. If $\tan \theta - \cot \theta = a$ and $\cos \theta + \sin \theta = b$, then $(b^2 - 1)^2 (a^2 + 4) = ?$

- a) 2 b) 1 c) 0 d) 4

43. $\tan^2 \alpha = 1 + 2 \tan^2 \beta$ then find the value of $\sqrt{2} \cos \alpha - \cos \beta = ?$

- a) 0 b) 1 c) 2 d) $\sqrt{2}$

44. If $2y \cos \theta = x \sin \theta$ and $2x \sec \theta - y \operatorname{cosec} \theta = 3$, then $x^2 + 4y^2 = ?$

- a) 1 b) 2 c) 0 d) 4

45. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ & $x \sin \theta - y \cos \theta = 0$ for every $\theta \in (0, \frac{\pi}{2})$, then what is $x^2 + y^2$ equal to?

यदि $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ & $x \sin \theta - y \cos \theta = 0$ प्रत्येक $\theta \in (0, \pi/2)$ के लिए, तो $x^2 + y^2$ किसके बराबर है?

(CDS 2023)

- A) 0
B) 1
C) 2
D) 3

46. If $\operatorname{Cosec} \theta - \sin \theta = 1$ & $\sec \theta - \cos \theta = m$, then $l^2 m^2 (l^2 + m^2 + 3) = ?$

- a) 0 b) 1 c) -1 d) 2

47. If $\sin \theta + \cos \theta = K$ and $\sec \theta + \operatorname{cosec} \theta = m$, find the value of $m(K^2 - 1)$.

यदि $\sin \theta + \cos \theta = K$ और $\sec \theta + \operatorname{cosec} \theta = m$ है, तो $m(K^2 - 1)$ का मान ज्ञात कीजिए।

- (a) $2M$ (b) K (c) $2K$ (d) M

48. $a(\tan \theta + \cot \theta) = 1$, $\sin \theta + \cos \theta = b$ and $0 < \theta < 90$, then relation between a and b ?

- a) $b^2 = 2(a+1)$ b) $b^2 = 2(a-1)$ c) $2a = b^2 - 1$ d) $2a = b^2 + 1$

49. Which of the following is true when $x = \sin A + \cos A$; $y = \sec A + \operatorname{cosec} A$?

$x = \sin A + \cos A$, $y = \sec A + \operatorname{cosec} A$ होने पर निम्नलिखित में से कौन-सा विकल्प सत्य होगा?

SSC CPO Pre 2024

- [A] $y(1 + x^2) = 2x$ [B] $y - 2x = x^2 y$
[C] $y + 2x = x^2 y$ [D] $y(1 - 2x^2) = x$



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50. If $\tan A + \sin A = p$ and $\tan A - \sin A = q$, then which of the following is true?

- a) $p^2 + q^2 = 4\sqrt{pq}$ b) $p + q = pq$ c) $p - q = pq$ d) $p^2 - q^2 = 4\sqrt{pq}$

51. If $a = \cot A + \cos A$ & $b = \cot A - \cos A$, then find the value of $a^2 - b^2 - 4\sqrt{ab}$?

यदि $a = \cot A + \cos A$ और $b = \cot A - \cos A$, तो $a^2 - b^2 - 4\sqrt{ab}$ का मान ज्ञात कीजिए?

- A) 0
B) -1
C) 1
D) -4

52. If $\cos^2 \theta - \sin^2 \theta = \tan^2 \phi$, then which of the following is true?

यदि $\cos^2 \theta - \sin^2 \theta = \tan^2 \phi$ है, तो निम्नलिखित में से कौन सा विकल्प सही है?

- (a) $\cos \theta \cos \phi = 1$ (b) $\cos^2 \phi - \sin^2 \phi = \tan^2 \theta$
(c) $\cos^2 \phi - \sin^2 \phi = \cot^2 \theta$ (d) $\cos \theta \cos \phi = \sqrt{2}$